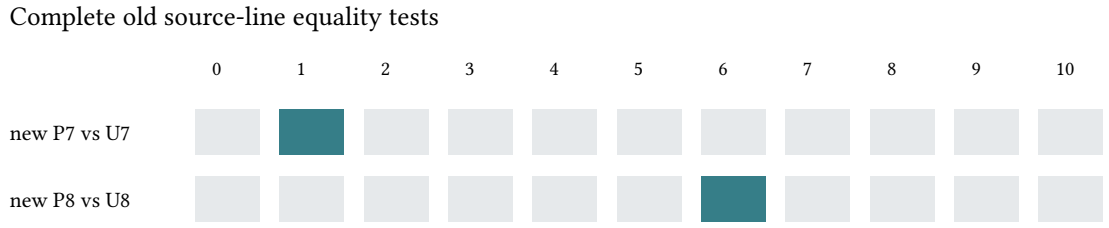
R48: exact source coordinates for the degree-13 class

Fixed-source observations under retained proper/RR/Picard interpretation. No independent verdict, chosen-target map or ICperformance comparison.



compare48-01: filled cells are exact matches; F=U7+6U8; chosen-target image unevaluated

Actual character action

Reusing the complete R47 degree-128 representations, only the missing two shifted signed sums and cofactors are computed. Both are nonprincipal and both isolated classes are killed by 11. The actual F,G=TF,H=T²F satisfy H+F=4G. Both P7=8F−G and P8=G−7F are nonprincipal, recover F and have actual diamond actions 7 and 8. No eigencharacter was assumed before testing.

Complete old-line comparison and source operators

Old R41 components U7,U8 have sealed input hashes and byte-identical global section data. Both nonzero and 11-annihilator controls are rechecked. All eleven multiples per line are compared. Unique matches are P7=U7 and P8=6U8, and native class addition independently reconstructs F=U7+6U8. Thus this new collected class is in the previously tested component span; no new direction outside it or independent target-image rank is established.

The full retained crossed operators reduce to 6F+2TF and 3F+9TF, match independent 9P7/9P8 calculations, and both images are nonprincipal. Scalar 5 on their sum recovers F. These are source endomorphism images. A chosen Koblitz-model isogeny or proper factor map remains a separate requirement.

Proof and failure custody

The new conditional Lean theorem gives uniqueness of coordinates below 11 for a cyclic line of exact order 11. Native arrays, curve normalization and Picard witness interpretation are not formalized. Existing R41 character proofs are reused by reference. Six native attempts pass. The first proof launcher failed before compiler start due to a module filename; its gate, old runner and stderr are retained. A corrected fresh attempt compiles and audits one theorem, zero failing or trusted declarations. No cap increased.

Next geometry task

The generic degree-32 source algebra is already known from R33. Next seek an actual normalization presentation, including registered boundary branches, using the installed positive-characteristic normalization library with its radical/component assumptions verified. This inspection is a new lead, not a completed normalization experiment. Proper charts/gluing, target identities, nonzero subgroup action and polarization remain open. Full integral recovery, reverse principal rows, saturation, theta/node187 and larger-n maps remain.

Details: PROTOCOL.md, COMPARE-PROTOCOL.md, FORMAT.md, results.json, LITERATURE.md. Fixed n=5 carrier; no attack or larger-n transfer claim.